Grind vs. Glory:

A Churn Analysis of College Football Retention

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Table of contents

Abstract 3

Introduction 4

Dataset & Preparation 5

Exploratory Data Analysis 16

Model & Feature Selection 20

Result 24

Conclusion 27

Synthesis & Discussion 27

Interpretability VS Prediction 28

Abstract

This study investigates the use of daily physical tracking metrics (Catapult data) to predict whether a college football player will enter the transfer portal. Because the dataset was limited to 71 players with a severe imbalance data, the Synthetic Minority Over-sampling Technique (SMOTE) and Stratified 5-Fold Cross-Validation were utilized to prevent model bias and overfitting. Through Forward Feature Selection, the dataset was optimized down to three key indicators: Average Player Load Per Minute, Season Max Velocity, and Home Latitude. This revealed that a player's physical intensity, peak athleticism, and geographic distance from home are significantly stronger predictors of transferring than bulk practice volume. Four machine learning algorithms—Logistic Regression, K-Nearest Neighbors, Naive Bayes, and XGBoost—were evaluated on this final feature set. XGBoost proved to be the definitive model, successfully capturing the complex, non-linear relationships in the data to achieve an 83% Accuracy, a 0.754 ROC/AUC, and a leading Precision of 0.677. Ultimately, this approach provides coaching staffs with a precise, actionable tool to identify risk of players' transfer enter while minimizing false alarms.

Introduction

In recent years, the NCAA, also known as National Collegiate Athletic Association, has faced dramatic changes that have had a significant impact on college football's recruiting, culture, and organization. These changes include unlimited transfer portal entry, allowing college athletes to transfer schools freely, and NIL compensations that include internal sponsors such as payments directly from the school organization, as well as external sponsors such public/private companies (ex: Gatorade, Nike, etc.). Here are some further information for those who aren't familiar.

- (NCAA1) - In April 2024 , the NCAA officially allowed unlimited transfers without penalty, followed by the U.S. District Court preliminary injunction and subsequent legal settlement.
- (NCAA2) - Effective in August 2024 , Division I Council approves NIL disclosure and transparency rules.
- (abc7) - Starting with the 2025-26 academic year, colleges can directly pay athletes. Due to the House v. NCAA settlement, schools can share up to $20.5 million annually (22% of average Power 4 revenue) directly with players, separate from NIL deals.

Prior to these changes, college athletes were guided by strict restrictions to go through the transfer process, including things such as having only one limited transfer portal enter, and a mandatory sit-out year (NCAA3). Additionally, NIL deals (Name Image Likeness deals), allows collegiate athletes to have a new revenue stream through third-party sponsors or directly from school organizations, were being completely banned after 2015, with historically being controversial all the time (Timeline of NIL). After these dramatic changes with regulations, because players no longer have to sit out a season when they switch schools, the sport has basically become year-round free

agency, and these college athletic organizations must understand how to operate like multi-million revenue generating businesses, rather than just some school organizations. For coaches, this means keeping the players you already have is just as important as recruiting new ones. Since programs spend massive amounts of time and money developing their athletes, figuring out which players might leave, before it happens, is a huge advantage.

This project will use classification models to predict why players enter the transfer portal. Although, getting players' financial data was not possible under the regulation of NCAA NIL deal being disclosed information, this project will concentrate the transfer factor in the performing side of analysis, rather than other external factors like financial issues. By comparing how hard a player works in practice, by using Catapult GPS data that I was able to access privately through the internship with the organization, and combine the usage of the players' playing time, recruiting information and other statistics on the field that is accessible in public through popular sports websites like ESPN. This study aims to find the exact breaking point where physical exhaustion and unmet expectations cause an athlete to transfer, and other factors that might potentially affect the players to look for different options for their career.

Dataset & Preparation

This project was aimed to gather datasets that could tell a story on the field in both the scenes, like the game environment, or behind the scenes like practices and other players' information to capture both the physical and situational factors that influence a player's decision to transfer. The first source is daily Catapult GPS data that captures metrics related to every session during season, which tracks physical exertion of athletes during practices, workouts, and games.

- roster_df - collected using the collegefootballdata.com API, which provides comprehensive data on college football teams, players, and activities. This dataset is essential for understanding the composition of the team and tracking player movements.

	weight	height	jersey	year	position	homeCity	homeState	homeLatitude	homeLongitude
0	181.0	71.0	25	4	DB	Oakland	CA	37.804456	-122.271356
1	174.0	71.0	4	4	WR	Compton	CA	33.894927	-118.226624
2	310.0	76.0	97	4	DL	Las Vegas	NV	36.167256	-115.148516
3	291.0	80.0	71	4	OL	Bethesda	MD	38.981985	-77.124234
4	231.0	75.0	84	4	TE	Visalia	CA	36.330228	-119.292058

- portal_df - also collected using the collegefootballdata.com API, specifically focusing on players who entered the transfer portal. This dataset is crucial for analyzing transfer patterns and predicting future transfers.

	position	destination	transferDate	rating	stars	eligibility
0	WR	Miami (OH)	2026-01-16T21:14:00.000Z	0.85	3.0	Immediate
1	OT	Georgia State	2026-01-13T00:13:00.000Z	0.85	3.0	Immediate
2	EDGE	Washington State	2026-01-08T19:52:00.000Z	0.86	3.0	Immediate
3	TE	UNLV	2026-01-13T17:34:00.000Z	0.86	3.0	Immediate
4	LB	Sam Houston	2026-01-07T17:01:00.000Z	0.85	3.0	Immediate

- catapult_df - collected through an internship with the Univ. of Arizona Football team, this dataset contains detailed player performance metrics from wearable technology. This dataset is valuable for analyzing player workload, performance trends, and potential injury risks, which can also influence transfer decisions.

	Player_Load	PL_Min	Distance	High_Speed_Distance	Hi_Accel	Hi_Decel	Max_V	Max_Vel
0	505.33	3.88	6376.32	58.26	11	10	20.66	
1	754.44	5.73	7423.43	34.54	17	6	20.60	
2	668.17	5.13	6104.08	26.73	25	12	20.40	
3	399.80	3.06	4705.43	55.13	6	2	20.40	
4	353.17	2.70	4029.53	15.22	4	6	20.34	

After collecting the datasets, I merged them into a single master dataset for modeling. The stat tables from the API combined easily because they already had one row per player. However, the Catapult data had multiple rows per player since it recorded daily practice sessions. To fix this, I grouped the Catapult data by player and aggregated their metrics before merging. Specifically, I added up their cumulative volume (total practice distance, player load, accelerations, and decelerations), calculated their average player_load_per_min, and extracted their seasonal best speed in mph.

I used an inner merge to only keep players who actually had Catapult data. Because the team has a limited number of GPS devices, some specialists (like kickers and punters) and non-starters do not wear them. As a result of this merge, only one player was dropped that has transferred. Additionally, after merging multiple datasets together, I also removed identification columns that are being repetitive and also to be confidential of players' information, such as their ID, jersey number, etc. Finally, the final dataset ended up being 97 rows of players, with 20 different columns. Here is a preview of the final master dataset before diving into the analysis.

- Total_Season_Distance: Sum of total distance (in yards) players moved during sessions.

	Total_Season_Distance
count	97.000000
mean	216814.232784
std	78713.749920
min	5403.340000
25%	170986.960000
50%	220904.520000
75%	274434.630000
max	358884.520000

- Total_Player_Load: Sum of composite measures of total physical exertion, calculated by Catapult's inertial measurement algorithm.

	Total_Player_Load
count	97.000000
mean	21426.298557
std	7796.630692
min	435.100000
25%	17706.160000
50%	21668.400000
75%	27017.670000
max	37670.650000

- Total_Contacts: Sum of contacts players took in every session recorded.

	Total_Contacts
count	97.000000
mean	760.123711
std	544.625947
min	0.000000
25%	338.000000
50%	570.000000
75%	1070.000000
max	2332.000000

- Total_Sessions: Total session recorded for specific player.

	Total_Sessions
count	97.000000
mean	54.051546
std	13.654495
min	3.000000
25%	52.000000
50%	60.000000
75%	63.000000
max	63.000000

- Avg_Player_Load_Per_Min: Average of all sessions player load per min.

	Avg_Player_Load_Per_Min
count	97.000000
mean	3.240673
std	0.712349
min	1.631207
25%	2.690397
50%	3.317162
75%	3.728571
max	4.874762

- High_Speed_Distance_Total: sum Total distance (in yards) covered by the athlete while moving at 80%+ of their maximum velocity.

	High_Speed_Distance_Total
count	97.000000
mean	1013.041443
std	770.483652
min	0.000000
25%	294.370000
50%	935.040000
75%	1604.490000
max	2780.240000

- Season_Max_Velocity: Fastest speed the players ran (in mph) during session.

	Season_Max_Velocity
count	97.000000
mean	19.280722
std	2.233227
min	12.180000
25%	18.540000
50%	19.910000
75%	20.810000
max	22.600000

- Total_High_Decels: Total count of high-intensity acceleration efforts for all the recorded session

	Total_High_Decels
count	97.000000
mean	131.237113
std	109.176823
min	0.000000
25%	26.000000
50%	107.000000
75%	215.000000
max	408.000000

- Total_High_Accels: Total count of high-intensity deceleration efforts for all the recorded session

	Total_High_Accels
count	97.000000
mean	240.731959
std	239.913331
min	0.000000
25%	30.000000
50%	189.000000
75%	394.000000
max	962.000000

- weight: Players' weight in pounds

	weight
count	97.000000
mean	232.721649
std	53.532456
min	161.000000
25%	191.000000
50%	220.000000
75%	265.000000
max	389.000000

- height: Players' height in inches

	height
count	97.000000
mean	73.907216
std	2.673491
min	69.000000
25%	72.000000
50%	74.000000
75%	76.000000
max	80.000000

- year: years of playing college football

```
1    32
4    26
3    25
2    14
Name: year, dtype: int64
```

- position: Players' position

```
['DB' 'OL' 'DL' 'RB' 'LB' 'QB' 'WR' 'TE']
```

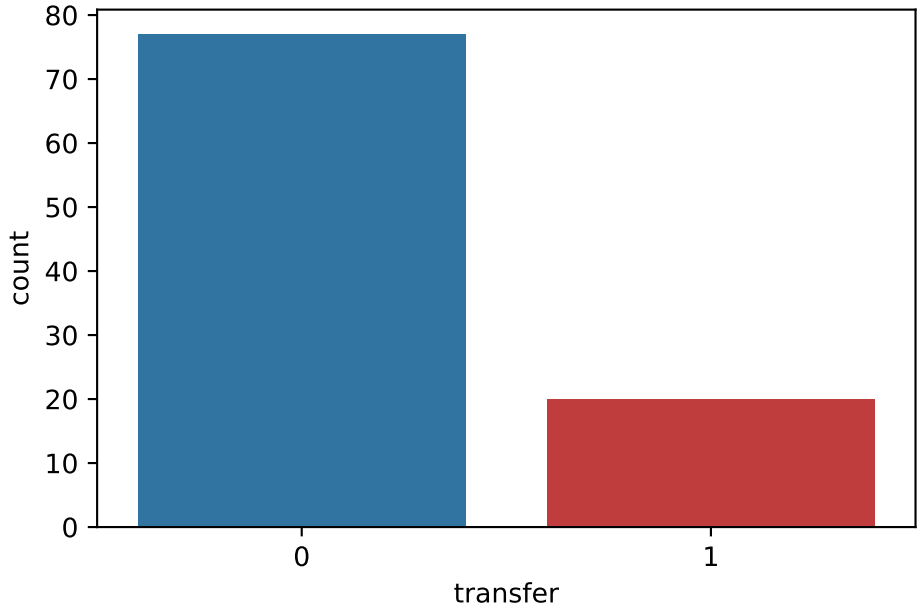
- homeCity: Players' home city
- homeState: Players' home state
- homeCountry: Players' home Country

- homeLatitude: Players' home Latitude
- homeLongitude: Players' home Longitude
- transfer: Binary indicator of whether player transferred after the 2025-2026 academic year season. (0 = Not transferred, 1 = Transferred)

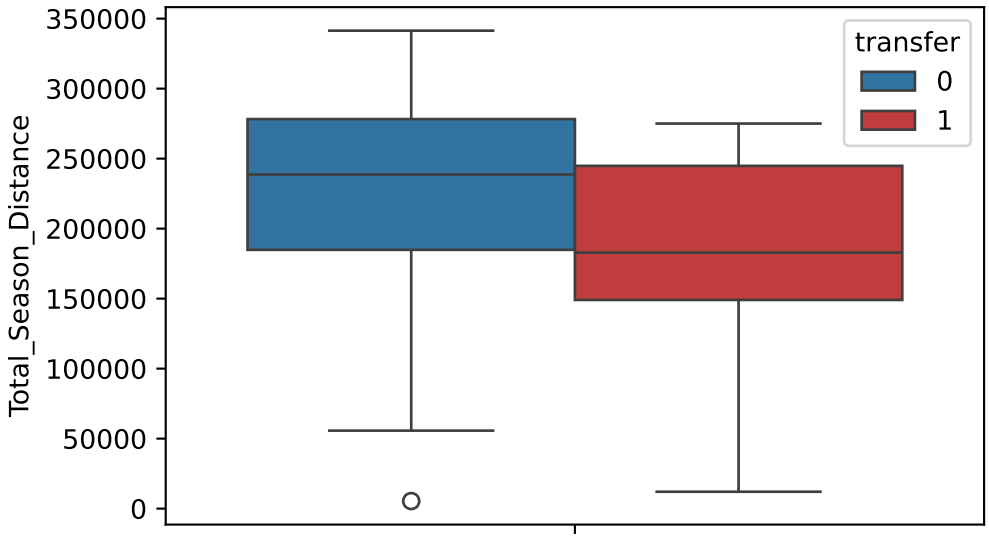
```
0    77
1    20
Name: transfer, dtype: int64
```

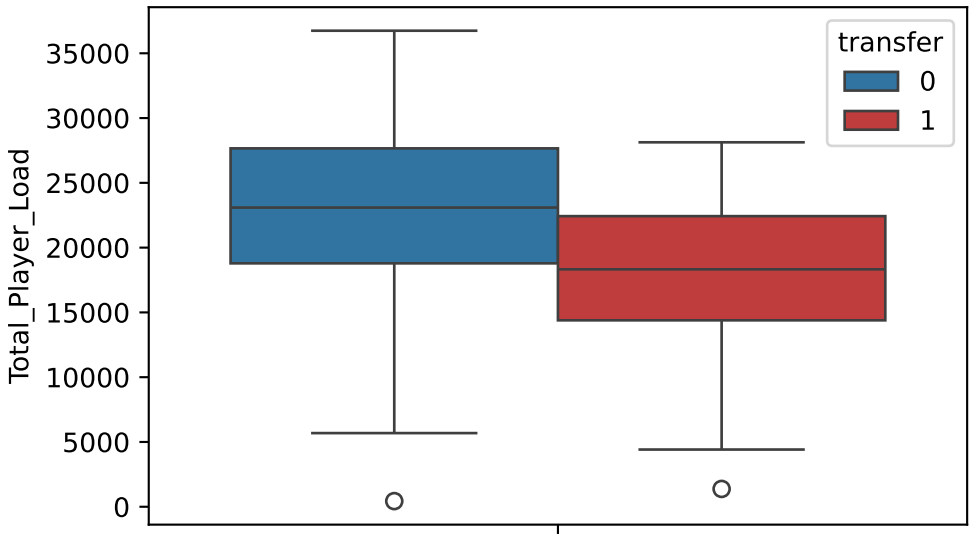
	0	1	2	3	4
Total_Season_Distance	249614.85	177899.69	213013.64	274434.63	334178.81
Total_Player_Load	21558.21	17921.91	22429.59	32182.12	28742.43
Total_Contacts	560	1568	1662	550	924
Total_Sessions	47	63	60	63	63
Avg_Player_Load_Per_Min	3.811064	2.329444	3.06725	4.232143	3.757063
High_Speed_Distance_Total	658.79	13.13	1724.67	1147.88	2005.72
Season_Max_Velocity	21.33	14.89	20.15	21.1	21.05
Total_High_Decels	207	6	103	106	266
Total_High_Accels	477	2	234	225	528
weight	185	331	234	192	176
height	71	79	77	71	72
year	1	2	3	3	4
position	DB	OL	DL	RB	DB
homeCity	Phoenix	Phoenix	Salt Lake City	Tucson	Philadelphia
homeState	AZ	AZ	UT	AZ	PA
homeCountry	USA	USA	USA	USA	USA
homeLatitude	33.448437	33.448437	40.75962	32.222876	39.952724
homeLongitude	-112.074142	-112.074142	-111.886798	-110.974848	-75.163526
transfer	0	0	1	0	0

Exploratory Data Analysis

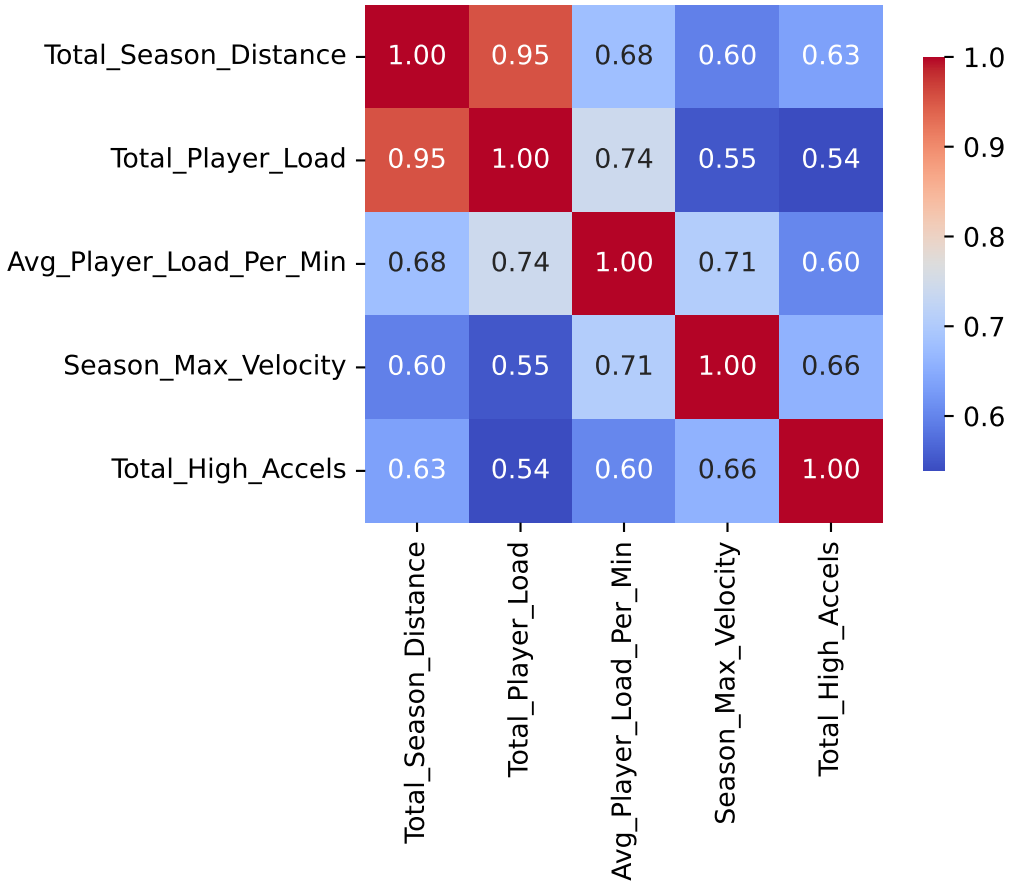


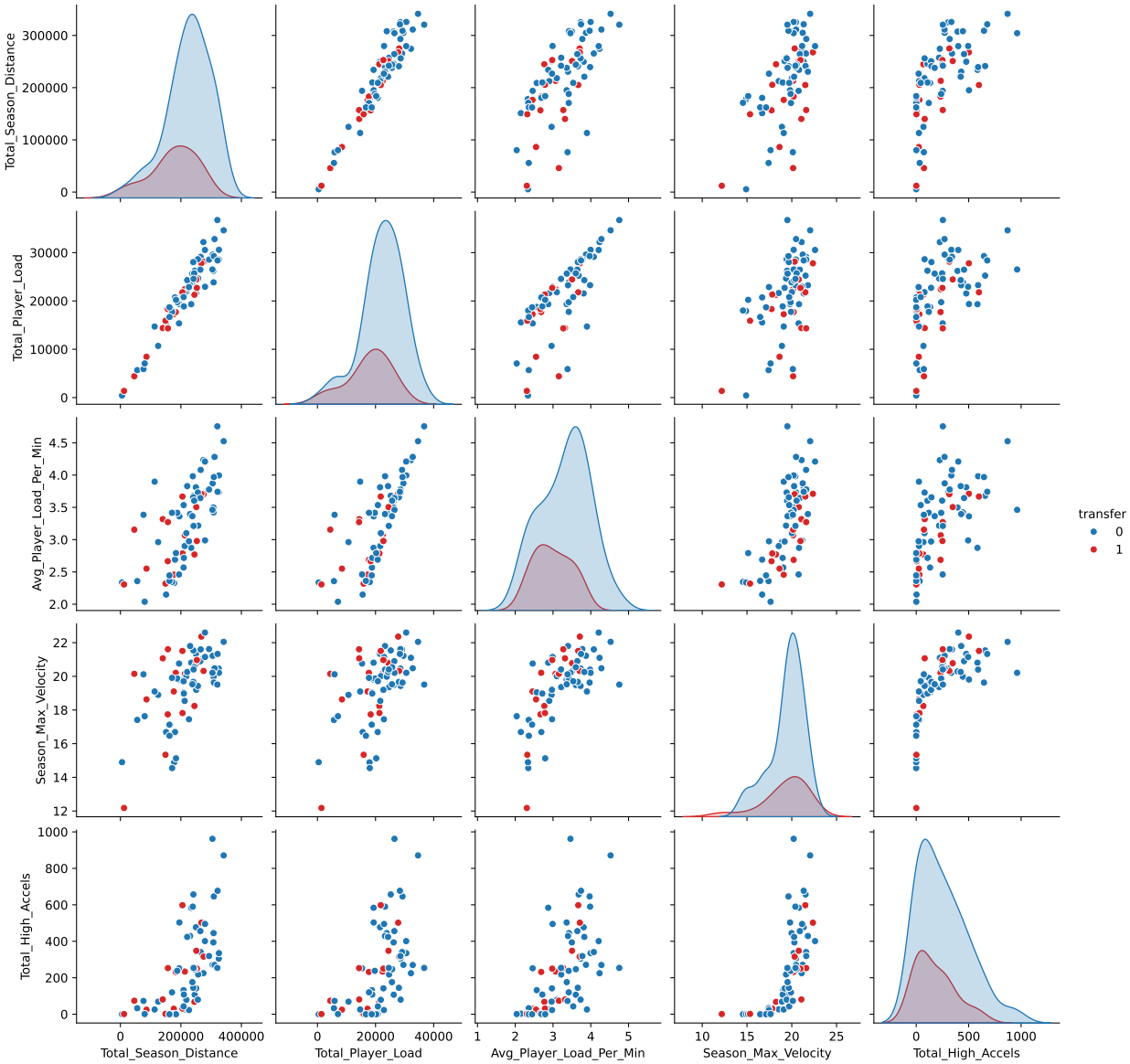
As we see, our response variable has imbalance data. Although thankfully it's less than half of players that are transferring, 22 is not a small number to ignore considering the fact that out of 77 players who did not transfer, 26 players were seniors which leaves us 51 players who did not transfer. By creating this bar plot, it also seems reasonable to remove all the seniors that will be out of eligibility year since they will not able to transfer no matter what.





After removing senior Football players, as we compare the boxplot of total distance and player load that was moved by players who transferred versus who did not, we can see that players who did not transfer had overall more distance moved and physical exertion than the other group. In a sport context, this plot makes sense, since the players who transferred are likely the players who did not get enough playing time during game, and out of the group who did not transfer, this group contains star players who gets to be on the field constantly, adding to the overall average to the non-transfer group. This column of data also contains wide range, but this is due to players who were injured during the season, unable to participate in practices or games whether the player transferred or not.


Total_Season_Distance
Total_Player_Load
Avg_Player_Load_Per_Min
Season_Max_Velocity
Total_High_Accels
1.00 0.95 0.68 0.60 0.63
0.95 1.00 0.74 0.55 0.54
0.68 0.74 1.00 0.71 0.60
0.60 0.55 0.71 1.00 0.66
0.63 0.54 0.60 0.66 1.00
1.0
0.9
0.8
0.7
0.6



To better understand the relationships within our features, I generated a correlation heatmap for the numerical Catapult metrics. Unsurprisingly, the visualization reveals strong positive correlations among the cumulative volume stats; for instance, Total_Season_Distance and Total_Player_Load are highly correlated since players who participate in more sessions naturally accumulate more load. We also see expected overlaps between explosive movement metrics, such as high accelerations and decelerations. Most importantly, the heatmap highlights how these physical tracking metrics correlate negatively with our target variable, transfer. Because several of

these GPS features provide redundant information, this heatmap helps identify which overlapping columns we can safely drop to simplify our machine learning model without losing predictive power.

To dive deeper into the relationships found in the heatmap, I generated a pair plot of the key Catapult metrics, color-coded by transfer status. This visualization allows us to see the exact distribution of each metric and how the features interact two at a time. By highlighting the players who transferred, we can look for distinct clusters. For example, if transferring players were consistently grouped together in the lower ranges of total distance and player load. While the heatmap showed us which variables mathematically overlap, the pair plot visually confirms if any combination of physical metrics creates a clear separation between players who stay and players who leave.

Model & Feature Selection

Given the limited sample size of 71 players and the severe class imbalance of the target variable, a standard single train-test split seems to create a significant risk. An 80/20 split could easily yield a test set containing zero transferring players, rendering model evaluation meaningless. While I considered Bootstrapping as a resampling alternative, its random sampling with replacement introduced the mathematical risk of generating training sets entirely devoid of the minority class. To effectively mitigate these issues, I implemented Stratified 5-Fold Cross-Validation. This approach separates the dataset into five distinct folds while strictly maintaining the original ratio of transfers to non-transfers in each split. Ultimately, this guaranteed that every player was utilized in the validation phase exactly once, providing a strong and stable evaluation of the models without risking the loss of critical minority class data.

For the feature selection, we will begin by removing all the string values, or other irrelevant information such as players' size, and also the response variable. Here are columns that will be dropped to fit the model.

```
['transfer', 'Name_x', 'position', 'homeCity', 'homeState', 'homeCountry', 'year', 'weig
```

With the dataset cleaned and the features selected, the next phase of this analysis involves determining the most effective machine learning algorithm to predict player transfers. The selected models include Logistic Regression, K-Nearest Neighbors (KNN), Gaussian Naive Bayes, and XGBoost. I began with Logistic Regression to establish a highly interpretable, linear baseline. I then trained the three more flexible, non-linear models like KNN, Naive Bayes, and XGBoost, to see if they could capture more complex, hidden relationships within the physical tracking metrics. By testing these four models simultaneously, the goal is to identify which specific architecture performs best at recognizing the profile of a transferring player.

```
--- 5-FOLD CV ACCURACY ---


Logistic Regression: 73.24%

K-Nearest Neighbors: 58.95%

Naive Bayes: 66.10%

XGBoost: 71.81%
```

Running the initial 5-Fold Cross-Validation revealed that our baseline Logistic Regression model performed the best, capturing an accuracy of 73.24%, closely followed by XGBoost at 71.81%. While these initial metrics seem promising, it is critical not to trust them right away. Because our

dataset of 71 players contains very few actual transfers, the models are inherently biased, which they can achieve these ~73% accuracies largely by ignoring the minority class and guessing that every player stays. To deliver more accurate results, I researched online solutions for handling small, imbalanced datasets and discovered the SMOTE (Synthetic Minority Over-sampling Technique) algorithm. SMOTE eliminates class bias by creating new, mathematically representative data points for the small data points, forcing the models to actually learn the profile of a transferring player. Moving forward, I will test all four models again using SMOTE to uncover their actual performance.

```
--- MODEL WITH SMOTE ---


Logistic Regression:
  Accuracy: 58.86%
  F1-Score: 0.393


K-Nearest Neighbors:
  Accuracy: 32.57%
  F1-Score: 0.239


Naive Bayes:
  Accuracy: 53.43%
  F1-Score: 0.373

XGBoost:
```

```
Accuracy: 71.81%

F1-Score: 0.433
```

Applying the SMOTE technique ended up help revealing the bias of the baseline model like logistic regression. When forced to train on a mathematically balanced dataset, the linear Logistic Regression model dropped to an accuracy of 58.86%, showing that its previous high score was largely a byproduct of guessing the majority class. K-Nearest Neighbors and Naive Bayes similarly struggled to find clear decision boundaries, resulting in poor predictive performance. However, the XGBoost algorithm demonstrated strong consistency and predictive power. It maintained a leading accuracy of 71.81% and achieved the highest F1-Score at 0.433. This proves that predicting a player's decision to transfer requires a flexible, non-linear model capable of capturing complex interactions between their physical Catapult metrics, rather than a simple linear equation. Based on these results, XGBoost was selected as the final model for this analysis, which I will now dive into what combination of features would deliver the best result.

To begin with a feature selection process, I needed to consider which subset selection would be most appropriate. Because our dataset is limited to 71 observations, feeding the model too many overlapping Catapult metrics risks overfitting and introduces unnecessary noise. To prevent this, I implemented Forward Feature Selection. Rather than starting with all variables (Backward Elimination), Forward Selection begins with an empty model and iteratively adds features only if they incrementally improve the cross-validated F1-Score by at least 1%. This steps ensure that the final model will ultimately be able to identify the primary indicators of a transferring player.

```
Best Features:

Index(['Avg_Player_Load_Per_Min', 'Season_Max_Velocity', 'homeLatitude'], dtype='object'
```

The Forward Feature Selection process successfully processed the dataset, revealing that the optimal predictive combination consisted of just three variables: Avg_Player_Load_Per_Min, Season_Max_Velocity, and homeLatitude. This completely changes how we look at the data. Instead of caring about bulk stats like total running distance or total practices, the XGBoost model focused on physical intensity and geography. By picking Avg_Player_Load_Per_Min and Season_Max_Velocity, the algorithm shows that how hard and fast a player works is a much bigger red flag for transferring than how long they practice. Adding homeLatitude to the mix shows that how far a player is from their hometown also plays a huge role in their decision, and if the players made a big decision to move all the way out here in Arizona, and not end up with the result they were hoping to get, such as becoming a starter or getting a good playing time after putting endless hours of work, this could discourage them significantly. Ultimately, this proves that predicting a college football transfer isn't just about total workload and financial issues, but it also comes down to a mix of explosive athleticism and where the player is from.

Result

After identifying the best model and feature, now it is time for reporting the result for our final model such as accuracy, precision, recall, and ROC/AUC curve.

```
--- Final Model Evaluation ---


Logistic Regression:

Accuracy: 0.659

Precision: 0.398
```

Recall: 0.650

F1-Score: 0.478

ROC/AUC: 0.651

KNN:

Accuracy: 0.702

Precision: 0.399

Recall: 0.683

F1-Score: 0.500

ROC/AUC: 0.712

Naive Bayes:

Accuracy: 0.521

Precision: 0.253

Recall: 0.567

F1-Score: 0.349

ROC/AUC: 0.648

XGBoost:

Accuracy: 0.830

Precision: 0.677

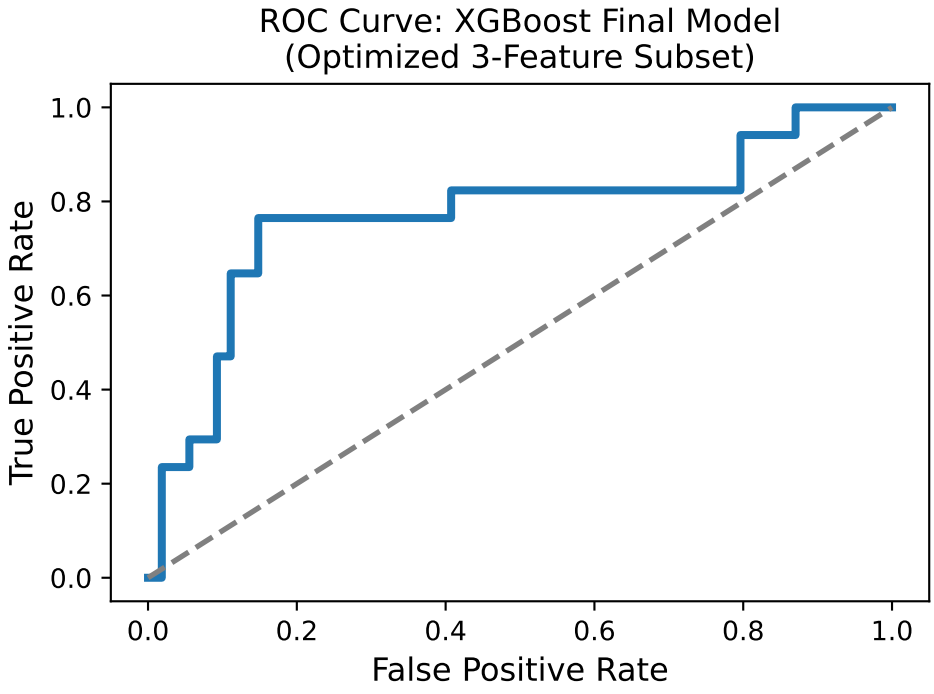
Recall: 0.617

F1-Score: 0.609

```
ROC/AUC: 0.754
```

Looking at the final scores for our three chosen stats, there is a clear winner. The simpler models, Naive Bayes and Logistic Regression, struggled the most. They couldn't figure out the complex patterns in the data and had the lowest overall scores. K-Nearest Neighbors (KNN) did a little better at actually catching the players who transferred (a Recall of 0.683), but it had one big mistake, which it generated too many false positive predictions. With a Precision of just 0.399, more than half of the players KNN flagged as transfers actually ended up staying.

XGBoost provided the most balanced and reliable performance. It achieved the highest overall Accuracy (83%) and the best ROC/AUC score (0.754). Most importantly, XGBoost had a massive lead in Precision (0.677) and F1-Score (0.609). This proves that XGBoost was highly capable of correctly identifying real transferring players without triggering a high number of incorrect guesses, making it the best model for this available dataset.



To visually demonstrate the predictive power of our final model, here is the ROC curve for the

XGBoost model. The dashed gray line on the graph represents a baseline of completely random guessing (an Area Under the Curve of 0.500). The blue line shows the performance of our XGBoost model. The higher and further left the blue line bows, the better the model is at catching real transfers without setting off false alarms. As shown in the figure, XGBoost achieves a strong curve with an AUC of 0.754. This visual confirms that the model successfully learned the patterns in the physical and geographic data, separating the transferring players from the retained players far better than random chance.

Conclusion

Synthesis & Discussion

The primary goal of this analysis was to determine if daily physical tracking data (Catapult metrics) could accurately predict whether a college football player would enter the transfer portal. Because the dataset was limited to 71 players with a severe imbalance of actual transfers, standard modeling approaches were easy to fail. By applying the SMOTE technique to balance the data and using Stratified K-Fold cross-validation, I created more precised testing environment. Ultimately, the XGBoost algorithm proved to be the most accurate model.

The most valuable takeaway from this project is the feature selection result. The model proved that stats that represents the volume of workload, like total running distance or the sheer number of practices, did not drive transfer decisions. Instead, the warning signs lie in a player's physical intensity (Avg_Player_Load_Per_Min), their peak athletic output (Season_Max_Velocity), and their geographic distance from home (homeLatitude). For a coaching staff, this is could be taken into

a actionable insight. It means coaches should be looking closely at their highly explosive, high-intensity athletes who are out-of-state, rather than just assuming the players who sit on the bench the longest are the ones who will leave.

Interpretability VS Prediction

When building machine learning models for a sports organization, there is always a trade-off between interpretability (how easily you can explain the math to a coach) and predictive power (how accurate the model actually is). A model like Logistic Regression is highly interpretable. It uses a simple linear equation, making it very easy to tell a coach exactly how much one specific stat impacts a player's chance of leaving. However, as this project showed, human behavior and physical performance does not always have a linear relationship. Logistic Regression failed to capture the complex patterns in the data and produced poor results.

On the other hand, XGBoost is a highly complex, non-linear model. It is often considered a "black box" because it is nearly impossible to explain the exact mathematical sequence of decision trees to coaches who have any mathematical background. However, it trades that simplicity for vastly superior prediction. In the environment where there are multiple million of dollars in line for college football roster management, prediction and result must be the priority. A coach ultimately needs to know who is at risk of leaving so they can intervene, and XGBoost was able to decently perform to identify those players with much higher precision and fewer false alarms. By extracting the top three driving features (Intensity, Velocity, and Geography), we successfully deliver evidence-based predictive analysis of a transfer churn analysis.

```
Trained model saved successfully!
```